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Subject : Research Methodology

Type : CCE II

Case Study:

A college student want to study the factors affecting students use of AI tools for academic work.

Q1) Answer the following questions

a) Define a suitable research problem for the given case:

A research problem refers to a difficulty faced by an individual or groups achieving certain objectives and requiring a systematic investigation to find a solution.

In this case the suitable research problem can be stated as:

To study and analyse the factors affecting students use of AI tools for academic work in the college.

- The college ^{want to} ~~student~~ understand students usage of AI tools.
- There are multiple possible factors influencing usage.
- There is a doubt regarding which factors are more significant.
- The study is limited to a specific college environment.

Thus it satisfies the components of a research problem.

b) Explain any two reasons why defining a research problem is necessary.

→ provide clear Direction.

A clearly defined problem helps the researcher identify what data to collect, which variables to study and what technique to use it avoids confusion and irrelevant data collection.

→ Helps in proper Research Design.

Defining the problem is prerequisite for preparing research design. It helps in formulating objectives, hypothesis and selecting appropriate methods.

An ill-defined problem creates hurdle in research.

c) Identify suitable primary and secondary data source for this study.

Primary Data Source:

primary data can be collected directly from students through:

Questionnaires.

Interviews.

Observation.

— Focus group discussion.

Secondary Data Source:

Secondary data can be obtained from:

Research articles on AI in education.

Books and academic journals.

Government or university reports.

College records.

secondary data must be checked for reliability, suitability and adequacy before use.

d) Explain the importance of literature review in the research:

Literature review is important because:

It provides background knowledge about AI tools in Education.

It helps identify gaps and limitations in previous studies.

It prevents duplication of research.

It helps refine the research problem and formulate objectives and hypotheses.

Thus literature review ensure that the research is relevant and scientifically sound.

e) Describe any two techniques involved in defining a research problem.

Statement of the problem in general way:

The problem should first be stated broadly. In case this case students use of AI tools for academic work is identified as the general problem.

Surveying the Available Literature:

The researcher should review existing studies, reports and theories related to AI in education. This helps narrow down the problem and define it closely in operation terms.